THE L^2 METRIC AT THE UHLENBECK BOUNDARY

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Abstract. We construct local Kähler potentials for the L^2 metric on moduli spaces of Hermitian–Yang–Mills connections over polarized projective surfaces that extend continuously to ideal connections. The potential combines a zeta determinant on a curve avoiding the bubble points with a weighted curvature integral and an explicit term for the defect cycle. On a projective K3 surface with a generic polarization and its Ricci-flat metric, let the Mukai vector be primitive, of rank at least two and positive square, with nonempty stable bundle locus. The potential identifies the L^2 metric with the canonical singular Ricci-flat metric on the normalized Uhlenbeck compactification. Its completion is the Uhlenbeck compactification, and in complex dimension $2N$ its volume is $(16\pi^4 \operatorname{Vol} X)^N/N!$. In rank two, the strata with stable residual bundle and distinct simple bubble points are totally geodesic hyperkähler orbifold strata, with the base-surface metric scaled by $4\pi^2$. At stable-residual boundary points, we identify the affine tangent cones and their Reeb actions and compute their volume densities from instanton partition functions.

1. Introduction

The L^2 metric on a moduli space of instantons records the size of infinitesimal changes of a connection. Its behaviour under bubbling is less direct: the curvature energy retains an atom at each bubble point, whereas the tangent vectors are defined on smooth connections. On a projective K3 surface this metric is hyperkähler, but the sheaf compactification and the Uhlenbeck compactification contract different geometric data. The question is which compact space carries its limiting metric, and how that metric restricts to the boundary.

We construct a local Kähler potential that is continuous at ideal connections. It combines a zeta determinant on a curve avoiding the bubble points with a logarithmically weighted curvature integral. Energy quantization supplies an explicit correction for the defect cycle. On K3 surfaces, this potential identifies the L^2 metric with the canonical singular Ricci-flat metric in its determinant class. Completion, volume and orbifold regularity then follow from the corresponding algebraic and analytic theorems.

1.1. **The potential at an ideal connection.** Let (X, ω) be a smooth projective Kähler surface with $[\omega] = c_1(H)$. We fix rank $r \geq 2$, determinant and Chern classes. For an ideal Hermitian–Yang–Mills connection (F, Z), choose a smooth curve $C \in |H^k|$ disjoint from $\operatorname{Supp} Z$, and a vector bundle G on C for which $\operatorname{End} F|_C \otimes G$ is acyclic. Such choices always exist. If $g_C = k^{-1} \log |s_C|$ and $\rho = \operatorname{rk} G$, our potential is

$$-\frac{\pi^2}{rk\rho} \log \det_\zeta \Delta_{\operatorname{End} F|_C \otimes G} - \frac{1}{2} \int_X g_C |F^0_{A_F}|^2 dV_X - 4\pi^2 \sum_x \operatorname{mult}_x(Z) g_C(x). \tag{1}$$

Date: November 13, 2026
This version: November 13, 2026.
2020 *Mathematics Subject Classification.* 53C07 (primary); 14D20, 14J28, 32Q20, 53C26 (secondary).
Key words and phrases. Instanton moduli, Uhlenbeck compactification, Kähler potential, Quillen metric, K3 surface, singular Ricci-flat metric, tangent cone.

Here $dd^c = (i/\pi)\partial\bar\partial$, the trace defining the metric is the fundamental trace, and $dd^c g_C = k^{-1}[C] - \omega$. The precise construction and coefficient calculation occupy Sections 2–3.

Theorem 1.1 (Continuous potential)**.** *The choices C, G can be fixed in a neighbourhood of every ideal connection. Formula (1) is continuous on that neighbourhood in the Uhlenbeck topology, and its complex Hessian on the smooth stable bundle locus is the L^2 Kähler form. On an obstructed moduli space the same identity holds after pullback to each smooth holomorphic family of stable bundles.*

The curve serves two purposes. Its determinant gives the tangential curvature term through the Quillen formula, and its logarithmic potential transgresses that term to the surface. Since the curve avoids the bubbles, the singular test function is integrated only where the curvature converges smoothly. Away from the curve, weak convergence of measures is sufficient. This separation is the reason a continuous potential can be obtained without estimates for instanton gluing parameters.

1.2. **The K3 consequences.** Let now X be a projective K3 surface with Ricci-flat Kähler form $\omega \in c_1(H)$. Let $v = (r, c, s)$ be primitive, $r \geq 2$, $v^2 > 0$, and let H be v-generic. Assume that the moduli space M° of slope-stable locally free sheaves is nonempty. Set $2N = v^2 + 2$ and $V = \mathrm{Vol}(X, g_X)$. Let M^ν be the normalization of the Uhlenbeck compactification; it is canonically homeomorphic to that compactification.

Theorem 1.2 (Canonical metric and boundary)**.** *The L^2 Kähler form extends to the canonical Ricci-flat current in an ample determinant class on M^ν, with continuous local potentials. Its metric completion is the Uhlenbeck compactification and*

$$\mathrm{Vol}(M^\circ, g_{L^2}) = \frac{(16\pi^4 V)^N}{N!}.$$

For rank two, along every stratum with stable residual bundle and ℓ distinct simple bubble points, the completed metric is a smooth hyperkähler orbifold metric. The stratum is totally geodesic, with intrinsic metric the product of the lower-charge L^2 metric and $\mathrm{Sym}^\ell(X, 4\pi^2 g_X)$ restricted to distinct points.

The current identification is proved in Theorems 4.2 and 5.1; the completion and boundary statements are Theorems 6.1 and 6.4. They also give finite curvature energy and an explicit lower bound for the diameter. A power of the Quillen metric extends continuously to an ample line on M^ν. In contrast, the L^2 metric cannot extend as a continuous positive-definite metric to the Gieseker moduli space when the Uhlenbeck morphism contracts a curve.

At a boundary point with stable residual bundle, we further identify the affine tangent cone and its Reeb grading with a product of affine ADHM instanton spaces. This determines its volume density by the corresponding Nekrasov coefficient (Theorems 7.1 and 7.3). Outside the rank-two distinct-point locus, identification of the entire tangent metric with the flat quotient metric is stated conditionally in Remark 7.2.

1.3. **Relation to earlier work.** Feehan [Fee95, Theorem 1.1] proves finiteness of the L^2 volume and diameter for irreducible ASD moduli spaces on closed, connected, oriented, simply connected four-manifolds with a generic Riemannian metric, assuming either $b^+ = 0$ with $G = \mathrm{SU}(2)$ or $\mathrm{SO}(3)$, or $G = \mathrm{SO}(3)$ with $w_2(P) \neq 0$. Under the same hypotheses, [Fee95, Theorem 1.2] states that the metric completion is homeomorphic to the Uhlenbeck compactification; the proof of that statement is deferred there to a separate work. These hypotheses and ours have different scopes. On a K3 surface the $\mathrm{SO}(3)$ case with $w_2(P) \neq 0$

is relevant, but the prescribed Ricci-flat metric need not satisfy Feehan's generic-metric hypothesis. Here the genericity assumption concerns the polarization for Gieseker stability. For primitive positive-square Mukai vectors with nonempty stable bundle locus, our argument applies in every rank $r \geq 2$ to the Ricci-flat representative itself. The continuous boundary potential identifies its L^2 metric with the canonical singular Ricci-flat metric; this gives the completion theorem in this setting, the exact volume, and the rank-two orbifold boundary geometry stated above.

Groisser's survey [Gro97] discusses the completion problem, and his analysis of totally geodesic boundaries [Gro98] establishes the scaled base metric in charge-one settings and discusses higher-charge strata. Our boundary formula retains the same factor $4\pi^2$ on the base surface, now directly from the curvature-energy atom.

Fibre-integral expressions for the generalized Weil–Petersson form and its determinant-line interpretation are established in [ST92, BS16]. Continuous potentials near locally free polystable bundles are obtained in [BuS21]. The new local ingredient here is the explicit potential at nonempty bubbling cycles, including its atomic term and its continuity proof. The algebraic–analytic compactification comparison is supplied by [BTT17, GSTW21]; we explain the fixed-polarization-class version used for the Ricci-flat representative. Once this potential is globalized in the determinant class, uniqueness in [EGZ09] identifies the metric. The completion theorem of [Son14] and the orbifold regularity theorem of [GPP25] then apply. The tangent-cone consequences use [NO26, DS17, LWX21]; these inputs are cited separately from the local potential argument.

The normalization of the metric matters throughout. We use $\|a\|_{L^2}^2 = -\int_X \operatorname{tr}(a \wedge *a)$, so a bubble has curvature energy $8\pi^2$. The rank-normalized Donaldson class is $-\frac{1}{2r} p_* \operatorname{ch}_2(\operatorname{End}\mathbb{E})$ paired with the surface class. These conventions determine both $4\pi^2$ in the boundary metric and $16\pi^4$ in the volume formula.

2. A local potential from a curve and the curvature density

Let X be a smooth projective surface, H an ample line bundle and $\omega = c_1(H, h_H)$ a Kähler form. We use the complex orientation and the conventions

$$dd^c = \frac{i}{\pi}\partial\bar{\partial}, \qquad c_1(H, h_H) = \frac{i}{2\pi} F_{h_H}, \qquad \operatorname{Vol}(X) = \frac{1}{2}\int_X \omega^2. \tag{2}$$

For a unitary connection A on a rank-r bundle E, write $F_A^0 = F_A - r^{-1}\operatorname{tr}(F_A)\,\mathrm{id}$. The determinant connection is fixed. Tangent vectors to the smooth moduli space of irreducible Hermitian–Yang–Mills connections are represented by trace-free skew-Hermitian one-forms in Coulomb gauge, and

$$g_{L^2}(a, b) = -\int_X \operatorname{tr}(a \wedge *b), \qquad \Omega_{L^2}(a, b) = g_{L^2}(Ja, b). \tag{3}$$

Here $(Ja)^{0,1} = ia^{0,1}$. On an obstructed moduli space all differential identities below are statements on its smooth stable locus, or after pullback to a smooth holomorphic family of stable bundles. The trace is the trace in the defining representation. For a Hermitian–Yang–Mills connection,

$$\int_X |F_A^0|^2\, dV_X = 8\pi^2\kappa, \qquad \kappa = c_2(E) - \frac{r-1}{2r} c_1(E)^2. \tag{4}$$

The trace-free curvature is primitive of type $(1,1)$ and hence anti-self-dual. One point bubble of charge one has curvature mass $8\pi^2$. These conventions fix the constants both in the potential and in the metric induced on a boundary stratum.

2.1. **The fibre integral.** Let $p : S \times X \to S$ and $q : S \times X \to X$ be the projections, where S is a complex manifold. Suppose that $\mathbb{E}$ is a holomorphic family of stable bundles with a smooth Hermitian metric that is Hermitian–Einstein on each fibre. Denote its classifying map by $f : S \to M^\circ$ and set

$$\Theta = \mathrm{ch}_2(\mathrm{End}\,\mathbb{E}, h).$$

All fibre integrals below use the complex orientation.

Lemma 2.1. *The pullback of the L^2 Kähler form is*

$$f^*\Omega_{L^2} = -\frac{2\pi^2}{r}\, p_*(\Theta \wedge q^*\omega). \tag{5}$$

Proof. Let F_0 be the trace-free curvature of the Chern connection on the family. The identity $\mathrm{tr}_{\mathrm{End}}(\mathrm{ad}\, a\, \mathrm{ad}\, b) = 2r\,\mathrm{tr}(a_0 b_0)$ gives

$$\Theta = -\frac{r}{4\pi^2}\,\mathrm{tr}(F_0 \wedge F_0).$$

For $v \in T_sS$ put $a_v = (\iota_v F_0)|_{X_s}$. The mixed curvature of a fibrewise Hermitian–Einstein family represents the harmonic Kodaira–Spencer class; this follows by differentiating the Hermitian–Einstein equation and using the Bianchi and Kähler identities [ST92, Proposition 1]. Thus a_v is the Coulomb representative of $df(v)$ and $a_{Iv} = Ja_v$. For real vectors v, w, contraction gives

$$\iota_w\iota_v\,\mathrm{tr}(F_0 \wedge F_0) = 2\,\mathrm{tr}\left(F_0(v,w)F_0\right) - 2\,\mathrm{tr}\left(\iota_v F_0 \wedge \iota_w F_0\right).$$

The first term vanishes after restriction to X_s and multiplication by ω, because $F_0|_{X_s} \wedge \omega = 0$. Consequently

$$p_*(\Theta \wedge q^*\omega)(v,w) = \frac{r}{2\pi^2}\int_X \mathrm{tr}(a_v \wedge a_w) \wedge \omega.$$

Pointwise, $\mathrm{tr}(a \wedge Ja) \wedge \omega = -|a|^2 dV_X$. Polarization identifies the integral on the right with $-\Omega_{L^2}(df(v), df(w))$, which proves (5). □

The fibre-integral description is classical; compare [BS16, Proposition 4.3]. The calculation above records its coefficient for the real metric (3).

2.2. **Transgression to an auxiliary curve.** Choose a smooth curve $C \in |H^k|$ and a section s_C defining it. Put

$$g_C = \frac{1}{k}\log|s_C|_{h_H^k}. \tag{6}$$

The logarithm is of the norm, not its square. Poincaré–Lelong gives

$$dd^c g_C = \frac{1}{k}[C] - \omega. \tag{7}$$

In particular, $g_C \in L^1(X)$ and $dd^c(-g_C) = \omega$ on $X \setminus C$.

Lemma 2.2. *If Θ is a smooth closed $(2,2)$-form on $S \times X$, then*

$$p_*(\Theta \wedge q^* dd^c g_C) = dd^c\left(\int_X g_C \Theta|_{X_s}\right). \tag{8}$$

The integral on the right is a smooth function of s.

Proof. Since Θ has pure type and is closed, $d^c\Theta = 0$. Multiplication of a distribution by a smooth form therefore gives $dd^c(q^*g_C\Theta) = q^*dd^c g_C \wedge \Theta$. Fibre integration commutes with ∂ and $\bar\partial$. The asserted smoothness follows by differentiating the smooth coefficients of Θ under the integral; on compact subsets of S, their derivatives are bounded and the fixed function g_C is integrable. □

On a Hermitian–Yang–Mills fibre,

$$\Theta|_{X_s} = -\frac{r}{4\pi^2}|F^0_{A_s}|^2 dV_X.$$

Substitution of (7) in (5) consequently yields

$$f^*\Omega_{L^2} = -\frac{2\pi^2}{rk}(p_C)_*(\Theta|_{S\times C}) + dd^c\left(-\frac{1}{2}\int_X g_C|F^0_{A_s}|^2 dV_X\right). \tag{9}$$

This is the step that replaces a potential on the surface by a determinant on a curve and a scalar integral on the surface.

2.3. **The determinant term.** Let G be a Hermitian holomorphic bundle of rank ρ on C, and suppose

$$H^j(C, \operatorname{End} E_s|_C \otimes G) = 0 \quad (j = 0, 1) \tag{10}$$

for $s \in S$. Fix a metric on C. Let $\Delta_s = \bar\partial^*\bar\partial$ on sections of $\operatorname{End} E_s|_C \otimes G$ with the induced metric. It is positive and invertible. Its zeta determinant is $\det_\zeta \Delta_s = \exp(-\zeta'_{\Delta_s}(0))$.

Lemma 2.3. *Under* (10)*,*

$$(p_C)_*(\Theta|_{S\times C}) = \frac{1}{2\rho}dd^c \log\det_\zeta \Delta_s. \tag{11}$$

Proof. Apply the curvature formula of Bismut–Gillet–Soulé to $\xi = \operatorname{End}\mathbb{E}|_{S\times C}\otimes q^*G$ [BGS88, Theorem 1.6, Remark 1.10 and equation (1.154)]. In the acyclic case the squared Quillen norm of the canonical determinant section is $\det_\zeta \Delta_s$, with

$$\frac{i}{2\pi}\partial\bar\partial \log\det_\zeta \Delta_s = \left[(p_C)_*\big(\operatorname{ch}(\xi)\operatorname{Td}(C)\big)\right]^{1,1}.$$

The Chern form $\operatorname{ch}_1(\operatorname{End}\mathbb{E}, h)$ vanishes identically. All forms involving G and $\operatorname{Td}(C)$ are pulled back from C. Thus the only term contributing a $(1,1)$-form on S is $\rho(p_C)_*\Theta$. The convention (2) gives the factor 2 in (11). □

Proposition 2.4. *On the locus* (10)*, the function*

$$u(A_s) = -\frac{\pi^2}{rk\rho}\log\det_\zeta \Delta_s - \frac{1}{2}\int_X g_C|F^0_{A_s}|^2 dV_X \tag{12}$$

is smooth, gauge invariant, and satisfies $dd^c u = f^*\Omega_{L^2}$.

Proof. Combine (9) and (11). Both the determinant and the curvature integral are invariant under unitary gauge transformations. Tensoring a universal family by a line bundle from S leaves its endomorphism bundle unchanged. Thus the same function is defined on the corresponding locus of gauge classes. □

The metric on $E_s|_C$ varies with s. In particular, positivity of a Quillen line for a fixed curve metric does not separately imply plurisubharmonicity of the first term of (12). Positivity is asserted for their sum, whose complex Hessian is (3).

For a Kähler form with $[\omega] = t\,c_1(H)$, $t > 0$, choose $c_1(H, h_H) = \omega/t$ and replace (6) by $g_C = (t/k)\log|s_C|$. The determinant coefficient in (12) becomes $-t\pi^2/(rk\rho)$; the curvature coefficient remains $-1/2$. Thus the formulas apply on any positive ray of a polarization with the expected scaling. We use $t = 1$ in the statements below.

3. Continuity at ideal connections

An ideal connection is a pair (A_F, Z), where F is a polystable locally free sheaf with its Hermitian–Yang–Mills connection and $Z = \sum_j n_j x_j$ is an effective zero-cycle. The determinant is fixed and $c_2(F) + \deg Z = c_2(E)$. We write M^{U} for the Uhlenbeck compactification of the chosen moduli component. Its algebraic structure is the reduced closure of that component in Li's projective slope-semistable compactification, as used in [Taj23, Section 7]. Only pairs in this closure are included.

3.1. **Algebraic and analytic convergence.** The comparison theorem needed here concerns the Kähler form used to solve the Hermitian–Yang–Mills equation, not only its cohomology class. We record the precise consequence.

Lemma 3.1. *Let $[\omega] = c_1(H)$, with any smooth Kähler representative. The algebraic and gauge-theoretic Uhlenbeck compactifications on a surface are homeomorphic. If $(F_i, Z_i) \to (F, Z)$, then, after unitary gauge changes, $A_{F_i} \to A_F$ smoothly on compact subsets away from* $\operatorname{Supp} Z$*, and*

$$|F^0_{A_{F_i}}|^2 dV_X + 8\pi^2 Z_i \rightharpoonup |F^0_{A_F}|^2 dV_X + 8\pi^2 Z \tag{13}$$

as measures. Here a zero-cycle denotes its atomic measure, with its algebraic multiplicities.

Proof. In rank two, continuity for flat families for arbitrary Gauduchon metrics is stated in [BTT17, Theorem 2.2]. For the arbitrary-rank surface homeomorphism and convergence assertion we adapt [GSTW21, Theorem 4.11 and Corollary 4.15] to the chosen representative. We describe the metric-dependent steps of this adaptation.

Replace H by a very ample power $L = H^a$, which does not change slope stability. The $\partial\bar\partial$-lemma supplies a metric on L with curvature $a\omega$. The bounded family of semistable sheaves admits a common regularity twist by L. For every fixed sufficiently high twist, choose a full L^2-orthonormal basis of sections. The Bochner inequality for an HYM bundle twisted by L^m gives $\Delta|s|^2 \le C_m|s|^2$ and therefore $\|s\|_\infty \le C'_m\|s\|_2$. These estimates use the curvature $a\omega$, not a Fubini–Study identity. After Uhlenbeck convergence, every basis section has a subsequence converging smoothly away from the bubbling set. The supremum estimate and the negligible volume of small neighbourhoods of that set imply

$$\langle s_{\infty,\alpha}, s_{\infty,\beta}\rangle_{L^2} = \delta_{\alpha\beta}.$$

Thus the whole basis survives, preserving the Hilbert-polynomial inequality needed to identify the image sheaf. The stabilization and full-rank arguments in [GSTW21, Lemmas 4.3–4.4] identify the Quot limit; Proposition 4.5 there identifies its slope-graded double dual. All slope comparisons involve the class $c_1(H)$ alone.

The individual multiplicities are compared by Chern–Weil transgression on local slices in [GSTW21, Propositions 4.7 and 4.9]. These steps use smooth Kähler geometry and local convergence, with no requirement that the metric be induced by the projective embedding. Writing E_0 for the algebraic torsion-free limit, they identify

$$F = (\operatorname{gr}^\mu E_0)^{**}, \qquad Z = \operatorname{cyc}\big(F/\operatorname{gr}^\mu E_0\big).$$

The corrected definition in [SW15, Erratum] assigns the generic associated-graded multiplicity, which on a surface is precisely the length of this zero-dimensional quotient; it is also the convention in [GSTW21, Section 2.5.3]. Finally, the compactness and diagonal argument of [GSTW21, Proposition 4.13 and Theorem 4.11] extends the comparison to sequences of ideal connections and descends it through the quotient. Corollary 4.15 gives the homeomorphism on surfaces. Uhlenbeck convergence then gives the smooth convergence and measure identity in the statement. □

3.2. **Auxiliary data near a boundary point.** Fix $p = (F, Z) \in M^{\mathrm{U}}$. A sufficiently high, general member $C \in |H^k|$ is smooth, avoids $\operatorname{Supp} Z$, and restricts each stable summand of F to a stable bundle [MR84]. Thus $F|_C$ is polystable, and $\operatorname{End} F|_C$ is semistable of degree zero. Faltings' criterion [Fal93, Theorem 1.2], in the form [BH08, Theorem 1.1], supplies a vector bundle G on C such that

$$H^*(C, \operatorname{End} F|_C \otimes G) = 0. \tag{14}$$

Choose any smooth Hermitian metric on G and a fixed metric on C. These choices depend on p; there is no requirement that one curve or one twist work on the whole compactification.

Choose a neighbourhood N of C with compact closure disjoint from $\operatorname{Supp} Z$. Shrinking around p in M^{U}, every nearby ideal cycle avoids $\overline{N}$, and the restricted Dolbeault operator remains invertible. Indeed, smooth convergence near $\overline{N}$ follows from Lemma 3.1. If atoms occurred in $\overline{N}$ along a converging sequence, a subsequence of their positions would converge there. Every such atom has mass at least $8\pi^2$, forcing an atom of that mass in the limiting measure, which is smooth on a neighbourhood of $\overline{N}$. This contradiction excludes the atoms. Invertibility is open in the smooth operator topology on the compact curve. We may consequently use the same C, G, N throughout an open neighbourhood U of p.

Theorem 3.2. *On U, define*

$$\begin{aligned} u(F', Z') = &-\frac{\pi^2}{rk\rho} \log \det_\zeta \Delta_{\operatorname{End} F'|_C \otimes G} - \frac{1}{2}\int_X g_C |F^0_{A_{F'}}|^2 dV_X \\ &- 4\pi^2 \sum_j n'_j g_C(x'_j), \qquad Z' = \sum_j n'_j x'_j. \end{aligned} \tag{15}$$

This function is continuous. Its restriction to every smooth holomorphic family of stable bundles in U is the potential of Proposition 2.4.

Proof. Let $(F_i, Z_i) \to (F', Z')$ in U. Choose a smooth cutoff χ supported in N and equal to one near C. On $\operatorname{Supp}\chi$, the curvatures converge smoothly modulo gauge. Since $g_C \in L^1(X)$,

$$\int_X \chi g_C |F^0_{A_{F_i}}|^2 dV_X \longrightarrow \int_X \chi g_C |F^0_{A_{F'}}|^2 dV_X. \tag{16}$$

The function $(1-\chi)g_C$ is continuous on all of X. Applying (13) to this function gives

$$\begin{aligned} &\int_X (1-\chi) g_C |F^0_{A_{F_i}}|^2 dV_X + 8\pi^2 \sum_x \operatorname{mult}_x(Z_i) g_C(x) \\ &\qquad \longrightarrow \int_X (1-\chi) g_C |F^0_{A_{F'}}|^2 dV_X + 8\pi^2 \sum_x \operatorname{mult}_x(Z') g_C(x). \end{aligned}$$

Here χ vanishes at every atom under consideration. Together with (16), this proves continuity of the last two terms in (15).

On the fixed curve, the operators converge smoothly and remain invertible. Their zeta determinants therefore converge [BGS88, Theorem 1.6 and Remark 1.10]. The determinant is gauge invariant, so the gauge changes used to obtain smooth convergence do not change its value. This proves continuity of u. On the locus of bundles the cycle is empty, and the assertion about its complex Hessian is Proposition 2.4. □

The two parts of the proof address different features of bubbling. The logarithmic singularity of g_C is harmless where curvature converges smoothly. Away from the curve the test function is continuous, so weak convergence of the curvature measures retains exactly the atomic term in (15). No rate of convergence and no gluing expansion are used.

The theorem gives a continuous function on the compactification. Positivity of its complex Hessian on the normalization, and its global cohomology class, require the arguments in Sections 4 and 5. At locally free polystable points it supplies a surface version of the continuous-potential extension of [BuS21, Theorem 11]; the additional term above treats nonempty bubbling cycles.

4. The compactification and its Kähler class

From this section onwards, X is a projective K3 surface and ω is the Ricci-flat representative of $c_1(H)$. Write

$$v = (r, c, s) = \operatorname{ch}(E)\sqrt{\operatorname{td} X}, \qquad (v, v) = c^2 - 2rs = 2N - 2.$$

We impose the following hypotheses throughout Sections 4–7: v is primitive, $r \geq 2$, $v^2 > 0$, H is v-generic for Gieseker stability, and the locus $M^\circ = M_H^\circ(v)$ of locally free μ_H-stable sheaves is nonempty. In particular $N \geq 2$. The determinant is fixed; on a K3 surface its isomorphism class is already fixed by c. The smooth projective Gieseker moduli space $M^{\mathrm{G}} = M_H(v)$ has complex dimension $2N$ and is of $\mathrm{K3}^{[N]}$ type [Yos99, Theorem 0.1]; see also [Yos01, Appendix, Theorem 8.1]. The open set M° is dense. Let $\gamma : M^{\mathrm{G}} \to M^{\mathrm{U}}$ be the Gieseker–Uhlenbeck morphism.

4.1. Normality and the local models.

We use the normalization of the image, denoted by M^ν, and the induced morphism $\pi : M^{\mathrm{G}} \to M^\nu$. The following description also identifies its points with ideal connections, without an ambiguity from normalization.

Proposition 4.1. *The morphism π is a projective symplectic resolution of a normal projective symplectic variety. Its fibres are connected, and $M^\nu \to M^{\mathrm{U}}$ is a homeomorphism. The complement of M° in M^ν has complex codimension at least two. At a point $(F, \sum_j n_j x_j)$ with F stable and the x_j distinct, its analytic germ is*

$$\mathbb{C}^{\operatorname{ext}^1(F,F)} \times \prod_j \mathfrak{M}_0(r, n_j), \qquad \operatorname{ext}^1(F,F) = 2N - 2r\sum_j n_j. \tag{17}$$

Here $\mathfrak{M}_0(r,n)$ is the affine ADHM quotient

$$\{(B_1, B_2, i, j) : [B_1, B_2] + ij = 0\} \,/\!/\, \mathrm{GL}(n), \tag{18}$$

where $B_\alpha \in \operatorname{End}\mathbb{C}^n$, $i : \mathbb{C}^r \to \mathbb{C}^n$, and $j : \mathbb{C}^n \to \mathbb{C}^r$. It includes its two-dimensional centre-of-mass factor.

Proof. Put $\beta_0 = (c \cdot H)/(rH^2)$. We use Tajakka's convention $\sigma = (\mathrm{Coh}^{\beta_0}(X)[-1], -Z_{\alpha,\beta_0})$ with $\alpha > 0$ [Taj23, Section 2.3]. Here $\mathrm{Coh}^{\beta_0}(X)$ is the usual tilt of coherent sheaves at slope

β_0, and Z is the central charge defined using the Chern character. The polystable object attached to an ideal connection is

$$F \oplus \bigoplus_j \mathcal{O}_{x_j}[-1]^{\oplus n_j}, \tag{19}$$

of Mukai vector v. This convention avoids a change of sign of v. On a K3 surface the unshifted charge is the usual Mukai central charge with parameter α' satisfying $(\alpha')^2 = \alpha^2 + 2/H^2$, followed by a positive rescaling of its imaginary part by α/α'. The heart is unchanged. Thus σ lies in the distinguished component of the stability manifold.

Let $U_H(v)$ denote Li's full reduced slope-semistable compactification, so that M^{U} is the reduced image of γ in $U_H(v)$. Tajakka's classification and comparison theorem [Taj23, Proposition 2.2 and Theorem 7.1] give a bijective morphism $U_H(v) \to M_\sigma(v)$, commuting with the same-H Gieseker morphisms. Its points record the slope-graded double dual and defect cycle; compare the Li–Gieseker–Uhlenbeck wall in [BM14, Proposition 8.2]. No auxiliary polarization is needed. The nonempty stable bundle locus gives stable objects, so this is not a totally semistable stability condition. The target is normal and irreducible by [AS23, Theorems 5.1 and 5.5]. The comparison morphism is projective and quasi-finite, hence finite. Its bijectivity and the irreducibility of the target imply that $U_H(v)$ is irreducible. The closed image M^{U} contains the nonempty open stable bundle locus, so $M^{\mathrm{U}} = U_H(v)$ as reduced spaces. The comparison is the identity on this common dense bundle locus. Its composition with the normalization of the reduced source is finite birational to a normal variety and is therefore an isomorphism. This also identifies that normalization with the Stein factor of γ, by their common function field and the proper birational map from M^{G}. Consequently π has connected fibres and $M^\nu \to M^{\mathrm{U}}$ is bijective. The latter is a homeomorphism by compactness.

A proper birational morphism onto a normal variety is an isomorphism outside a subset of complex codimension at least two in the target. Transfer the Mukai two-form from M^{G} to this open subset of M^ν_{reg}. Hartogs extension of its coefficients gives a holomorphic two-form on all of M^ν_{reg}. Its top wedge cannot vanish only in codimension at least two, so it is nondegenerate. Its pullback agrees with the Mukai form on a dense open subset of $\pi^{-1}(M^\nu_{\mathrm{reg}})$ and hence everywhere there. The Mukai form on M^{G} is consequently its nondegenerate extension to the resolution. This proves that π is a symplectic resolution. Equality of the two nondegenerate forms also makes $d\pi$ invertible over M^ν_{reg}, so π is an isomorphism there. In particular M^ν is symplectic, canonical and has trivial canonical class, and π is crepant.

For completeness, the codimension assertion can be read directly in the local quiver description [AS23, Proposition 3.1 and Corollary 4.1]. Let $\mathbf{m}$ be the local dimension vector and let $p(\mathbf{a}) = 1 - \frac{1}{2}(\mathbf{a}, \mathbf{a})$ for the doubled-quiver Euler form. Since simple objects occur densely, $\mathbf{m} \in \Sigma_0$. For every proper representation type $\mathbf{m} = \sum_t k_t \boldsymbol{\beta}_t$, the simple-root criterion gives

$$p(\mathbf{m}) > \sum_t k_t p(\boldsymbol{\beta}_t).$$

The stratum of this type has dimension $2\sum_t p(\boldsymbol{\beta}_t)$ [CB01, Theorems 1.2–1.3]; its codimension is therefore at least two. A stable phase-one object of the positive rank under consideration is a slope-stable locally free sheaf. Indeed a point defect supplies a point factor in (19). Thus the nonsimple locus is exactly the complement at issue.

Finally suppose F is stable. Its vertex has dimension one and $\mathrm{ext}^1(F, F)/2$ loop pairs. The vertex at x_j has dimension n_j, two loops, and r arrows in each direction connecting it to the F-vertex. There are no arrows between distinct point vertices. The scalar moment-map equation at the F-vertex follows by taking the traces of the remaining equations; the diagonal

scalar acts trivially. Eliminating these redundant scalars leaves exactly (18) for each j and the free loop coordinates at F. Riemann–Roch gives $\operatorname{ext}^1(F,F) = (v+\ell(0,0,1))^2+2 = 2N-2r\ell$, where $\ell = \sum_j n_j$. This proves (17). □

4.2. **The ample determinant class.** On the bundle locus the universal endomorphism bundle defines its usual Chern character. To extend this class over M^{G}, use the derived endomorphism class of the universal sheaf. Explicitly, if a quasi-universal sheaf $\mathbb{E}$ has similitude m, set

$$\delta_v = m^{-2}\operatorname{ch}_2(R\mathcal{H}om(\mathbb{E},\mathbb{E})) = m^{-2}\bigl(2rm\operatorname{ch}_2(\mathbb{E}) - c_1(\mathbb{E})^2\bigr).$$

This is the discriminant class, independent of twisting by a line from the base, and equals $\operatorname{ch}_2(\operatorname{End}\mathbb{E})$ for an actual universal bundle on M°. On the boundary, ordinary sheaf endomorphisms are not a substitute for this derived class. For $\alpha \in H^2(X,\mathbb{R})$ define

$$\mu_v(\alpha) = -\frac{1}{2r}p_*(\delta_v\, q^*\alpha). \tag{20}$$

The familiar coefficient $-1/4$ is the rank-two instance of (20). Put

$$\Lambda = 4\pi^2\mu_v([\omega]) \in H^2(M^{\mathrm{G}},\mathbb{R}). \tag{21}$$

The determinant construction of the Uhlenbeck morphism gives an ample rational class on M^ν whose positive real multiple Λ' satisfies $\pi^*\Lambda' = \Lambda$. More explicitly, for a sufficiently high curve $D \in |H^k|$, the determinant class λ_D satisfies

$$c_1(\lambda_D) = -\frac{k}{2}p_*(\delta_v\, q^*c_1(H)), \qquad \Lambda = \frac{4\pi^2}{rk}c_1(\lambda_D). \tag{22}$$

A positive power of λ_D descends to an ample line bundle on M^ν [HL10, Section 8.2]. These formulas are interpreted rationally when the moduli space is not fine.

Theorem 4.2. *There is a unique closed positive current T on M^ν with continuous local potentials whose restriction to M° is Ω_{L^2}. Its class is Λ'. It can be written $T = \theta + dd^c\varphi$, where θ is a smooth representative of Λ' and φ is continuous on M^ν.*

Proof. Pull the functions of Theorem 3.2 to M^ν. They are continuous and plurisubharmonic on its dense open set M°. The removable-singularity theorem for locally bounded plurisubharmonic functions on normal spaces extends this property to M^ν [GR56, Sätze 3–4]. On overlaps, a difference is continuous and pluriharmonic on M°. Applying the same extension theorem to it and its negative shows that it is pluriharmonic everywhere. The complex Hessians therefore glue to T. This argument also proves uniqueness among continuous-potential extensions.

Let $\widetilde{T} = \pi^*T$. Its class and Λ agree on M° by Lemma 2.1. The kernel of restriction $H^2(M^{\mathrm{G}},\mathbb{R}) \to H^2(M^\circ,\mathbb{R})$ is generated by the classes of the irreducible divisors in $M^{\mathrm{G}} \setminus M^\circ$. This follows from the cohomology sequence with supports and the Thom class at each generic smooth point of a divisor; subsets of complex codimension at least two contribute nothing in degree two. All these divisors are exceptional by Proposition 4.1. Hence $[\widetilde{T}] - \Lambda = \sum_i a_i[D_i]$.

The restriction of a local potential for $\widetilde{T}$ to any fibre of π is constant. Its class has degree zero on every contracted curve; the same holds for Λ. The exceptional real divisor $\sum_i a_iD_i$ is thus numerically trivial over M^ν. Apply the negativity lemma to this divisor and its negative [KM98, Lemma 3.39]; both are relatively nef, so all a_i vanish.

The $\partial\bar\partial$-lemma on the compact Kähler manifold M^{G} now gives $\widetilde{T} = \pi^*\theta + dd^c\widetilde{\varphi}$. The potential is continuous, since locally it differs from the explicit continuous potential by a smooth function and a pluriharmonic function. It is constant on each compact connected

fibre: locally along such a fibre its complex Hessian is zero, and a pluriharmonic function on a connected compact complex space is constant. Properness gives a continuous descended function φ. This proves the global assertion. □

5. The canonical singular Ricci-flat metric

5.1. **Identification by the Monge–Ampère equation.** Let σ denote the holomorphic symplectic form on M^ν_{reg}, normalized so that its restriction to M° is the form supplied by hyperkähler reduction. Since π is a symplectic resolution, this form extends nondegenerately to M^{G}.

Theorem 5.1. *The current T of Theorem 4.2 is the canonical Ricci-flat current in the ample class Λ'. It is a smooth hyperkähler metric on M^ν_{reg}, and its restriction to M° is g_{L^2}. Moreover,*

$$\mathrm{Vol}(M^\circ, g_{L^2}) = \frac{(16\pi^4 \mathrm{Vol}(X, g_X))^N}{N!}. \tag{23}$$

Proof. The L^2 metric is hyperkähler [Ito88, Theorem 2.1]. Consequently, on M° its volume form satisfies

$$\Omega_{L^2}^{2N} = b\,(\sigma \wedge \overline{\sigma})^N$$

for a positive constant b, with the conventional sign absorbed in the positive measure on the right. The left side is the Bedford–Taylor Monge–Ampère measure of a continuous-potential current. It gives no mass to proper analytic subsets. The symplectic measure on the right also gives no mass to the boundary: on the resolution it is a smooth volume form. The identity therefore holds on all of M^ν, in the sense of measures. The cohomology class fixes b by integration.

The space M^ν is projective, canonical and has trivial canonical class. The existence and uniqueness theorem of Eyssidieux–Guedj–Zeriahi [EGZ09, Theorem 7.5 and Corollary 7.6] applies to the ample real algebraic class Λ'. Its unique bounded-potential solution is T. The same theorem makes T smooth on M^ν_{reg}. The parallel symplectic structure agrees there with that on the dense open set M°, so this regular metric is hyperkähler.

Bounded potentials have full cohomological mass; hence

$$\mathrm{Vol}(M^\circ, g_{L^2}) = \frac{1}{(2N)!}\int_{M^{\mathrm{G}}} \Lambda^{2N}. \tag{24}$$

Let q be the Beauville–Bogomolov form normalized by the Mukai isometry. The vector associated to $\mu_v(\alpha)$ is, up to sign, $(0, \alpha, c\cdot\alpha/r) \in v^\perp$. It has square α^2. The Mukai isometry [OGr97, Yos99] thus gives

$$q(\Lambda) = 16\pi^4 \int_X \omega^2 = 32\pi^4 \mathrm{Vol}(X, g_X).$$

The Fujiki relation for $\mathrm{K3}^{[N]}$ type is

$$\int_{M^{\mathrm{G}}} \Lambda^{2N} = \frac{(2N)!}{2^N N!}\, q(\Lambda)^N.$$

Substitution in (24) proves (23). □

The right side of (23) is also the volume of $\mathrm{Sym}^N(X, 4\pi^2 g_X)$. This equality is an intersection-theoretic identity; it does not assert an isometry with that symmetric product.

5.2. **The Quillen metric.** The preceding argument also controls a natural determinant metric. For a large positive integer k, put

$$\mathcal{L} = \big[\det Rp_*(R\mathcal{H}om(\mathbb{E},\mathbb{E}) \otimes (H^k - H^{-k}))\big]^{-1}. \tag{25}$$

This expression is read on the bundle locus for its Quillen metric and in perfect K-theory for its global determinant class. A fixed trivial-rank term may be included in this virtual bundle without affecting its varying curvature. Let m be the similitude of the chosen quasi-universal sheaf. The actual determinant line retains the factor m^2 in its curvature and class; positive powers clear rational descent data, not this factor.

Corollary 5.2. *On M°, a positive power of the Quillen metric on $\mathcal{L}$ is, up to a constant factor, the restriction of a continuous Hermitian metric on a positive power of an ample line bundle on M^ν.*

Proof. Grothendieck–Riemann–Roch and Quillen curvature give

$$c_1(\mathcal{L}, h_Q) = -2km^2\, p_*(\delta_v\, \omega) = \frac{rkm^2}{\pi^2}\Omega_{L^2}; \qquad c_1(\mathcal{L}) = 4m^2 c_1(\lambda_D).$$

These are the surface determinant formulas of [BS16, Theorem 4.6]. The current T gives a continuous metric on the corresponding ample line downstairs, after multiplication by its positive constant and passage to powers. On M° the ratio of the two metrics is a flat unitary Hermitian line bundle.

This remaining flat line has finite monodromy. Indeed $H^1(M^{\mathrm{G}}, \mathbb{Q}) = 0$. The degree-one cohomology sequence for the complement of the boundary injects $H^1(M^\circ, \mathbb{Q})$ into the vector space generated by its divisors, with next map sending a divisor to its class in $H^2(M^{\mathrm{G}}, \mathbb{Q})$. Those divisors are exceptional, and their classes are linearly independent: a cohomologically trivial real exceptional divisor is relatively numerically trivial and vanishes by the negativity lemma, as in Theorem 4.2. Hence $H^1(M^\circ, \mathbb{Q}) = 0$. The group $H_1(M^\circ, \mathbb{Z})$ is finitely generated, so it is finite. A further power trivializes the unitary character by a parallel section. The metrics then agree up to a constant. □

The vanishing used in this proof is specific to the present K3 moduli space. Equality of curvatures alone would leave a possibly infinite unitary character on a general surface moduli space. We make no such continuous Quillen-extension assertion there.

5.3. **Why the Gieseker space does not carry the extended metric.**

Corollary 5.3. *If π contracts a curve, g_{L^2} has no continuous positive-definite Riemannian extension to M^{G}.*

Proof. Suppose such an extension exists. Since the complex structure is fixed, continuity makes the extension Hermitian, with a continuous positive $(1,1)$-form $\widetilde{\Omega}$. It is closed as a current. To see this, remove tubular neighbourhoods of the analytic boundary and use cutoffs: a bounded form paired with the gradient of a cutoff near a real codimension-two set gives an error tending to zero; higher-codimension pieces are smaller. Closedness on M° therefore extends across the boundary. Locally a continuous closed $(1,1)$-form has a continuous potential w; strict positivity makes w strictly plurisubharmonic. On the complement of the boundary, $w - \pi^* u$ is pluriharmonic, where u is one of the potentials of Theorem 3.2. Continuity and removability extend that identity across the boundary. On any contracted curve, $\pi^* u$ is constant. Thus w restricts locally to a pluriharmonic function on that curve, contradicting strict plurisubharmonicity. □

This distinguishes the existence of a hyperkähler metric on the smooth Gieseker moduli space from extension of its particular L^2 metric. Tyurin's extension assertion [Tyu92, Theorem 2], made for a fine component, cannot hold as an extension of this positive-definite metric when the Uhlenbeck morphism contracts a curve. Example 8.2 below is fine and has such curves. The obstruction is intrinsic to the class Λ, which is zero on every contracted curve.

6. Completion and the geometry of boundary strata

6.1. Completion and finite curvature energy.

Theorem 6.1. *The metric completion of (M°, g_{L^2}) is canonically homeomorphic to M^{U}. It is the compact length space defined by the canonical metric on M^ν. It is a noncollapsed* $\mathrm{RCD}(0, 4N)$ *space, with its Riemannian volume measure. If η is a Kähler class on M^{G}, the Ricci-flat metrics in $\Lambda + \varepsilon\eta$ converge to this completion in the Gromov–Hausdorff topology as $\varepsilon \downarrow 0$, and converge smoothly on compact subsets over M^ν_{reg}.*

Proof. The class Λ' is a positive multiple of an ample rational class and π is a crepant resolution. Song's theorem [Son14, Theorem 1.1 and Corollary 1.1] identifies the completion of the regular metric with M^ν, proves compactness and geodesic convexity of its regular set, and bounds the Hausdorff dimension of the singular set by $4N - 4$. The additional closed analytic set $M^\nu_{\mathrm{reg}} \setminus M^\circ$ has real codimension at least four. Removing it does not change intrinsic distances: a piecewise smooth curve in the regular locus can be perturbed, with arbitrarily small length change, to avoid an analytic subset of real codimension at least two. It follows that completing M° gives the same space. Proposition 4.1 identifies its topology with M^{U}.

The convergence assertion is [CT15, Theorem 1.6], applied to the nef and big class Λ; see also [RZ11, Theorems 1.1–1.2]. The approximating metrics are Ricci-flat and their volumes tend to the positive number in (23). Their compact Gromov–Hausdorff limit is therefore noncollapsed. Stability of the synthetic Ricci lower bound and volume convergence give the stated RCD structure [DPG18]. Here noncollapsed means that the reference measure is the $4N$-dimensional Hausdorff measure, with its usual Euclidean normalization. □

Corollary 6.2. *Writing $V = \mathrm{Vol}(X, g_X)$ and $D = \mathrm{diam}(M^{\mathrm{U}})$, one has*

$$D \geq (16\pi^2 V)^{1/4} \left(\frac{(2N)!}{N!} \right)^{1/(4N)} \geq \left(\frac{64\pi^2 N V}{e} \right)^{1/4}. \tag{26}$$

The curvature is square integrable on M°. More precisely, define $|R|^2_{\mathrm{op}} = \frac{1}{4} \sum_{i,j,k,l} R^2_{ijkl}$ in a real orthonormal frame. Then

$$\int_{M^\circ} |R|^2_{\mathrm{op}} dV_{g_{L^2}} \leq \frac{48\pi^2(N+3)}{(N-1)!} (16\pi^4 V)^{N-1}. \tag{27}$$

Proof. Bishop–Gromov comparison gives $\mathrm{Vol}(M^{\mathrm{U}}) \leq \omega_{4N} D^{4N}$, where $\omega_{4N} = \pi^{2N}/(2N)!$. Substitution of (23) gives the first bound. Integrating $\log x$ over $[N, 2N]$ yields $(2N)!/N! \geq (4N/e)^N$, giving the second.

For a compact Ricci-flat Kähler manifold of complex dimension $2N$, the Chern–Weil identity in this norm is

$$\int |R|^2_{\mathrm{op}} dV = \frac{8\pi^2}{(2N-2)!} \int c_2\, \Omega^{2N-2}$$

[HS01, formula following (8)]. Apply it to the approximating metrics in Theorem 6.1 and use smooth convergence on compact subsets and exhaustion. The right side tends to its value at Λ. For $\mathrm{K3}^{[N]}$ type,

$$\int c_2 \alpha^{2N-2} = 6(N+3)(2N-3)!!\, q(\alpha)^{N-1}.$$

This coefficient follows as well by comparing the degree $N-1$ terms in Riemann–Roch and $\chi(L) = \binom{q(c_1(L))/2+N+1}{N}$. Using $q(\Lambda) = 32\pi^4 V$ gives (27). □

6.2. **Restriction of the current to a stratum.** Put $v_\ell = v + \ell(0,0,1)$ and let $M_H^\circ(v_\ell)$ denote the stable locally free locus, when nonempty. On the symmetric product, let $\omega_X^{(\ell)}$ be the current whose pullback to X^ℓ is $\sum_{a=1}^{\ell} \mathrm{pr}_a^* \omega$.

Theorem 6.3. *There is a holomorphic map*

$$j_\ell : M_H^\circ(v_\ell) \times \mathrm{Sym}^\ell X \longrightarrow M^\nu, \qquad (F, Z) \longmapsto (F, Z),$$

which is injective on points. Its pullback current is

$$j_\ell^* T = \Omega_{L^2, v_\ell} + 4\pi^2 \omega_X^{(\ell)}. \tag{28}$$

This identity includes cycles with repeated points.

Proof. Work over a local universal family of stable bundles F. The relative Quot scheme of length-ℓ quotients of F maps to M^{G} by taking kernels and to $\mathrm{Sym}^\ell X$ by their cycles. The kernels are slope stable. They belong to the unique irreducible Gieseker moduli space M^{G}, hence to the closure of M°. Fibres over a cycle are products of punctual Quot schemes and are connected by [EL99, Theorem 1]. Composition with π is constant on each such fibre. The base $M_H^\circ(v_\ell) \times \mathrm{Sym}^\ell X$ is normal; proper descent therefore gives the required holomorphic map. These local constructions agree, including when the universal families differ by a line from the base. The description of points gives injectivity.

Fix (F, Z) and choose C, G as in Section 3. In (15), the first two terms depend only on F and have Hessian Ω_{L^2, v_ℓ}. On a neighbourhood disjoint from C the last term has Hessian

$$dd^c \left(-4\pi^2 \sum_a n_a g_C(x_a) \right) = 4\pi^2 \sum_a n_a \omega(x_a).$$

This proves (28) on the ordered product; the symmetric expression descends, also across its diagonals. All potentials are continuous, so their pullbacks and Hessians are defined there. □

The current identity alone does not identify an induced Riemannian metric on a singular stratum. The next theorem supplies that extra regularity in rank two with distinct bubble points.

6.3. **Rank two and distinct points.** Let $r = 2$, and let S_ℓ be the image of the locus with stable F and ℓ distinct points, each of multiplicity one.

Theorem 6.4. *Near S_ℓ, the completed metric is a smooth hyperkähler orbifold metric. The stratum is totally geodesic, and its intrinsic metric is*

$$g_{L^2, v_\ell} + 4\pi^2 g_X^{(\ell)}. \tag{29}$$

At $p \in S_\ell$ the metric tangent cone and normalized volume density are

$$\mathbb{R}^{4N-4\ell} \times (\mathbb{R}^4/\{\pm 1\})^\ell, \qquad \vartheta(p) = 2^{-\ell}. \tag{30}$$

In equivariant normal coordinates on the covering chart, the metric coefficients differ from their values at the centre by $O(\rho^2)$.

Proof. The ADHM space of one rank-two instanton is $\mathfrak{M}_0(2,1) = \mathbb{C}^2 \times (\mathbb{C}^2/\{\pm 1\})$. Equation (17) gives the local quotient

$$\mathbb{C}^{2N-2\ell} \times (\mathbb{C}^2/\{\pm 1\})^\ell.$$

The orbifold regularity theorem of Guenancia–Pan–Păun [GPP25, Corollary B] says that the canonical Ricci-flat metric is smooth on the whole quotient-singularity locus. Its pullback to the covering chart is thus a smooth Ricci-flat metric. The parallel symplectic form and complex structures extend over the fixed loci, so the metric is hyperkähler there as well.

The stratum lifts to the common fixed-point locus of the isometric group $(\mathbb{Z}/2)^\ell$. A fixed-point locus of isometries is totally geodesic. Its metric is (29), since its Kähler form is the restriction computed in Theorem 6.3. The invariant tangent-space decomposition gives (30); the finite quotient divides Euclidean volume by 2^ℓ. Finally the first derivatives of a smooth metric vanish in normal coordinates. The exponential map is equivariant, giving the stated quadratic coefficient estimate. □

In particular, if the lower moduli space in S_1 satisfies the hypotheses of Theorem 5.1, then

$$\operatorname{Vol}(S_1) = \frac{(16\pi^4 V)^{N-1}}{(N-2)!}. \tag{31}$$

Indeed its complex dimension is $2N-4$, and the product in (29) has volume

$$\operatorname{Vol}(M_H^\circ(v_1)) \operatorname{Vol}(X, 4\pi^2 g_X).$$

The same calculation applies if the lower space consists of a single spherical bundle, with volume one. Equation (31) concerns S_1. It equals the $(4N-4)$-dimensional measure of the entire singular locus only if its complement there has measure zero in that dimension. Reducible residual bundles require a separate analysis; odd first Chern class alone does not supply that conclusion.

6.4. **A Lipschitz estimate for curvature measures.** The following estimate does not require a complex structure on the base.

Proposition 6.5. *Let X be a compact oriented Riemannian four-manifold and let M° be a smooth moduli space of irreducible projectively anti-self-dual connections with fixed determinant and energy $8\pi^2\kappa$. For every Lipschitz function f on X, the function*

$$\Phi_f(A) = \frac{1}{2}\int_X f|F_A^0|^2 dV_X$$

is $2\pi\sqrt{\kappa}\,\operatorname{Lip}(f)$-Lipschitz in the L^2 metric. In the K3 setting it extends to the completion by

$$\Phi_f(F, Z) = \frac{1}{2}\int_X f|F_{A_F}^0|^2 dV_X + 4\pi^2 \sum_j n_j f(x_j). \tag{32}$$

In particular, two one-bubble points with the same background obey

$$d_{M^\cup}((F,x),(F,y)) \geq \frac{2\pi}{\sqrt{\kappa}} d_X(x,y). \tag{33}$$

Proof. For smooth f and a harmonic tangent vector a, the Yang–Mills equation and integration by parts give

$$d\Phi_f(a) = -\int_X \langle \iota_{\nabla f} F_A^0, a\rangle dV_X.$$

An anti-self-dual two-form satisfies $|\iota_w F_A^0|^2 = \frac{1}{2}|w|^2|F_A^0|^2$. Cauchy–Schwarz therefore bounds the absolute value by $2\pi\sqrt{\kappa}\|df\|_\infty\|a\|_{L^2}$. The same integration by parts holds for a Lipschitz f using its weak first derivatives, or follows by local smooth approximation. Integration along paths proves the asserted Lipschitz constant. The completion theorem and weak convergence of the energy measures give (32). For equal backgrounds its difference is $4\pi^2(f(x) - f(y))$. Taking the supremum over all 1-Lipschitz functions proves (33). □

For the rank-two stratum with a fixed stable background, Theorem 6.4 also gives the upper bound $d_{M^{\mathrm{U}}}((F,x),(F,y)) \le 2\pi d_X(x,y)$ by paths inside the stratum. For sufficiently close points in a convex orbifold neighbourhood this upper bound is an equality. Global total geodesicity does not by itself make every long intrinsic geodesic globally minimizing.

7. Tangent cones and instanton densities

The local model (17) is an analytic statement. Identifying its natural grading with the grading of a metric tangent cone requires an additional argument. We give that argument before discussing metrics on the resulting cone.

7.1. The affine cone and its Reeb action.

For an affine cone, the Reeb action is the circle action associated to its positive grading. We normalize it so that a holomorphic volume form in complex dimension d has weight d. On (17), the standard action multiplies all linear coordinates, including B_1, B_2, i, j, by the same scalar. The holomorphic symplectic form has weight two, so this action has the stated normalization.

Theorem 7.1. *Every point of M^ν has a unique metric tangent cone. Its affine complex variety is the conical symplectic germ associated to that point. If $p = (F, \sum_j n_j x_j)$ has stable residual bundle F, then this affine variety, with its normalized Reeb action, is*

$$\mathcal{C}_p = \mathbb{C}^{\mathrm{ext}^1(F,F)} \times \prod_j \mathfrak{M}_0(r, n_j) \tag{34}$$

with the standard action just described.

Proof. The projective symplectic resolution in Proposition 4.1 satisfies the hypotheses of [NO26, Theorems 1.1 and 6.1]. That result identifies the canonical conical symplectic germ with the affine variety of the metric tangent cone. Its polarized smoothing permits the uniqueness theorem of [DS17, Theorem 1.3] to be applied. Neither assertion requires the singularity to be isolated.

It remains to identify the Reeb action at a stable-residual point. Endow (34) with the zero-level flat hyperkähler quotient metric. Its squared radius is a continuous proper homogeneous function on the affine quotient, by the Kempf–Ness identification. Relative to an algebraic homogeneous radius its logarithmic ratio is bounded, since a link is compact. It is a Kähler potential on the free locus and extends plurisubharmonically across the singular set.

Here the free locus is the whole regular locus. One way to check the necessary simple-root condition is to use the extended ADHM quiver of dimension vector $(1, n)$. Simple representations exist for $r \ge 2$: take B_1 diagonal with distinct eigenvalues, $B_2 = 0$, let the first two columns of i be a vector a with all entries nonzero, and let the first two rows of j be $b, -b$, where all entries of b are nonzero. Then $ij = 0$. No proper coordinate subspace

contains $\operatorname{im} i$, and no nonzero coordinate subspace lies in $\ker j$. Thus the representation is simple. The simple-root criterion [CB01, Theorem 1.2] and the smooth-locus theorem [BSc21, Theorem 1.15] identify regular points with simple representations, equivalently free points modulo the diagonal scalar. The assertion for the product follows factor by factor.

On this regular locus the quotient metric is hyperkähler and satisfies the Ricci-flat Monge–Ampère equation. Its potential and volume weight therefore satisfy the definition of a weak Ricci-flat cone metric in [LWX21, Definitions A.6–A.8]. Corollary A.4 of that paper makes the standard cone K-polystable. By [LX18, Theorem 3.5], its Reeb valuation minimizes normalized volume. The minimizer is unique up to scaling [NO26, Theorem 2.4]; normalizing the volume-form weight removes that scalar. The stable degeneration theorem, in the form recalled in [NO26, Section 2], identifies the first associated graded cone with (34). It is already K-polystable, so the unique K-polystable degeneration of [LWX21, Theorem 1.2] makes no further change. This is the grading of the metric tangent cone. □

Remark 7.2. Theorem 7.1 identifies the affine cone and its Reeb action. To identify its metric with the flat ADHM quotient metric in full generality, we would also need the following uniqueness assertion:

> (Q) A Ricci-flat Kähler cone metric with locally bounded potential on a K-polystable cone is unique, for a fixed Reeb action, up to an automorphism commuting with that action.

Under (Q), the two cone metrics in question are isometric. We retain this implication as conditional, in view of the general uniqueness issue described in [NO26, Remark 6.14]; compare [DS17, Proposition 4.8] and [LWX21, Theorem A.12]. No such additional assertion is needed in Theorem 6.4, where the tangent metric is explicitly flat upstairs. It is also unnecessary for the volume densities below.

7.2. **Densities from the Hilbert character.** For a graded affine variety C, write $H_C(q) = \sum_{m\ge 0} \dim \mathbb{C}[C]_m q^m$ for its Hilbert character. The normalized metric density at p is

$$\vartheta(p) = \lim_{\rho\downarrow 0} \frac{\operatorname{Vol}(B_\rho(p))}{\omega_{4N}\rho^{4N}}.$$

The limit exists by volume comparison.

Theorem 7.3. *At an ideal point $p = (F, \sum_j n_j x_j)$ with stable residual bundle F,*

$$\vartheta(p) = \prod_j \theta(r, n_j), \qquad \theta(r,n) = \lim_{t\downarrow 0} t^{2rn} H_{\mathfrak{M}_0(r,n)}(e^{-t}). \tag{35}$$

The factor $\theta(r,n)$ is the n-instanton coefficient of the $U(r)$ Nekrasov partition function at $\varepsilon_1 = \varepsilon_2 = 1$ and $a_1 = \cdots = a_r = 0$, with specialization performed after summing and cancelling the rational localization expression. In particular,

$$\theta(r,1) = 4^{-(r-1)} \binom{2r-2}{r-1}. \tag{36}$$

Proof. Noncollapsed volume convergence identifies $\vartheta(p)$ with the vertex density of the tangent cone. The leading Hilbert-character coefficient of a normalized Ricci-flat cone is its volume ratio [HeS17, Appendix C]. By Theorem 7.1, the character is that of (34). Characters multiply under products; the Euclidean factor contributes one. This proves (35) without requiring a product decomposition of the tangent metric.

The Gieseker resolution $\mathfrak{M}(r,n) \to \mathfrak{M}_0(r,n)$ has a rational-singularities target. Its equivariant Euler characteristic of the structure sheaf is therefore the character of the affine

coordinate ring. The leading term of equivariant Riemann–Roch is the equivariant integral of one, computed in [NY05, Proposition 4.1 and (4.6)]. This is the instanton coefficient in the statement. Standard linear scaling of the ADHM data agrees on the quotient with the torus specialization $\varepsilon_1 = \varepsilon_2 = 1$, $a = 0$: the two lifts differ by a central $\mathrm{GL}(n)$ action. Individual fixed-point terms may have poles there; their sum, interpreted as this positively graded equivariant multiplicity, has the asserted finite value.

For $n = 1$ the quotient is $\mathbb{C}^2$ times the minimal nilpotent orbit closure in $\mathfrak{sl}_r$. The coordinate ring of the minimal orbit closure in degree $2k$ has dimension

$$\dim V_{k\vartheta} = \frac{2k + r - 1}{r - 1} \binom{k + r - 2}{r - 2}^2,$$

where ϑ here denotes the highest root. The leading coefficient of this polynomial in k is $2/((r-1)((r-2)!)^2)$ and its degree is $2r - 3$. Since $\sum_{k\geq 0} k^m e^{-2kt} \sim m!/(2t)^{m+1}$, the leading character coefficient is

$$\frac{2(2r-3)!}{(r-1)((r-2)!)^2\, 2^{2r-2}} = 4^{-(r-1)} \binom{2r-2}{r-1},$$

as required. □

For reproducibility, the rational expression used for the higher values is as follows. Let $\mathbf{Y} = (Y_1, \ldots, Y_r)$ range over tuples of Young diagrams with total size n. For a box $s = (i, j)$, put $A_Y(s) = Y_i - j$ and $L_Y(s) = Y'_j - i$, allowing negative values when s is outside the diagram. Define

$$\begin{aligned} E_{\alpha\beta}(\mathbf{Y}) = &\prod_{s\in Y_\alpha} [a_\beta - a_\alpha - \varepsilon_1 L_{Y_\beta}(s) + \varepsilon_2 (A_{Y_\alpha}(s) + 1)] \\ &\cdot \prod_{s\in Y_\beta} [a_\beta - a_\alpha + \varepsilon_1 (L_{Y_\alpha}(s) + 1) - \varepsilon_2 A_{Y_\beta}(s)]. \end{aligned}$$

Then

$$Z_n(\varepsilon_1, \varepsilon_2; a) = \sum_{|\mathbf{Y}|=n} \prod_{\alpha,\beta=1}^{r} E_{\alpha\beta}(\mathbf{Y})^{-1} \tag{37}$$

is the tangent-weight formula of [NY05, Theorem 2.11]. For example, with $r = 2$ and $a_1 = a$, $a_2 = -a$, writing $e = \varepsilon_1 + \varepsilon_2$ gives

$$Z_2 = \frac{8e^2 + \varepsilon_1\varepsilon_2 - 8a^2}{\varepsilon_1^2\varepsilon_2^2 (e^2 - 4a^2)((2\varepsilon_1 + \varepsilon_2)^2 - 4a^2)((\varepsilon_1 + 2\varepsilon_2)^2 - 4a^2)}.$$

Substitution gives $\theta(2, 2) = 33/324 = 11/108$. The same finite sums give Table 1. The accompanying exact-arithmetic calculation evaluates a generic one-parameter approach before taking its limit, so it never assigns a value to an individual singular summand.

TABLE 1. Normalized densities of the affine instanton cones.

(r, n)	$(2, 1)$	$(2, 2)$	$(2, 3)$	$(3, 1)$	$(3, 2)$
$\theta(r, n)$	$1/2$	$11/108$	$61/5184$	$3/8$	$61/1296$

The rank-one value $\theta(1,n) = 1/n!$, obtained from $\operatorname{Sym}^n \mathbb{C}^2$, is a normalization check outside the rank assumption of the main theorem. The inequalities

$$\frac{11}{108} < \frac{1}{4}, \qquad \frac{61}{5184} < \frac{11}{216}, \qquad \frac{61}{1296} < \left(\frac{3}{8}\right)^2$$

are consistent with lower semicontinuity of metric density when points collide. Neither the product formula nor this table describes reducible residual bundles: their full Ext quivers replace (34).

8. Examples and extensions

8.1. **Two moduli spaces on a double plane.** Let $\phi : X \to \mathbb{P}^2$ be a double cover branched over a smooth sextic, with $\operatorname{Pic}(X) = \mathbb{Z}H$ and $H = \phi^*\mathcal{O}_{\mathbb{P}^2}(1)$. Then $H^2 = 2$ and $\operatorname{Vol}(X) = 1$ in our normalization. Put

$$F_0 = \phi^* T_{\mathbb{P}^2}(-1).$$

The Euler sequence gives $\operatorname{rk} F_0 = 2$, $c_1(F_0) = H$, $c_2(F_0) = 2$ and $v(F_0) = (2, H, 1)$, of square -2.

We verify the stability needed in the examples. A destabilizing saturated rank-one subsheaf would extend to a nonzero map $\mathcal{O}_X(aH) \to F_0$ with $a \geq 1$. Since $\phi_*\mathcal{O}_X = \mathcal{O} \oplus \mathcal{O}(-3)$,

$$H^0(X, F_0(-aH)) = H^0(\mathbb{P}^2, T_{\mathbb{P}^2}(-1-a)) \oplus H^0(\mathbb{P}^2, T_{\mathbb{P}^2}(-4-a)) = 0$$

by the Euler sequence. Thus F_0 is stable. It is spherical: simplicity and Serre duality give $\operatorname{ext}^1(F_0, F_0) = v(F_0)^2 + 2 = 0$. It is also the unique stable sheaf of its Mukai vector. A second, nonisomorphic stable sheaf of that vector would have no homomorphisms in either direction, whereas Riemann–Roch would give Euler characteristic two, a contradiction.

Example 8.1. Take $v = (2, H, 0)$. Then $N = 2$, $c_2 = 3$ and $\kappa = 5/2$. Every semistable sheaf is slope stable, since equality of the rank-one slope with $H^2/2$ would require an integer multiple of H with intersection $H^2/2$. If a sheaf has defect length ℓ, its stable double dual has square $2 - 4\ell \geq -2$, so $\ell \leq 1$. The non-locally-free locus is the projective bundle of one-dimensional quotients of F_0 over X, of dimension three. Kernels of these quotients exist, and their moduli space is smooth of dimension four. Its bundle locus is therefore nonempty. The Uhlenbeck morphism contracts each $\mathbb{P}^1$ fibre and adds precisely one copy of X to M°.

The completion has volume $128\pi^8$. Its boundary is totally geodesic with metric $4\pi^2 g_X$, and at every boundary point the tangent cone is $\mathbb{R}^4 \times (\mathbb{R}^4/\{\pm 1\})$, of density $1/2$. This Gieseker moduli space is generally not fine: the algebraic Mukai pairings have common divisor two. The conclusions above do not require a universal sheaf on the coarse space.

Example 8.2. Take $v = (2, H, -1)$. Then $N = 4$, $c_2 = 4$ and $\kappa = 7/2$. Stable kernels of quotients $F_0 \to T$ of length two give nonempty Gieseker moduli. The stable residual bundles have either vector $(2, H, 0)$ and one bubble, or vector $(2, H, 1)$ and two bubbles. The latter residual bundle is F_0. The length-one non-locally-free locus in the Gieseker space has dimension $4 + 2 + 1 = 7$, including its $\mathbb{P}^1$ Quot fibre. The length-two locus has dimension six. Both are smaller than eight, so the bundle locus is nonempty. The Uhlenbeck boundary consists of

$$M_H^\circ(2, H, 0) \times X \quad \text{and} \quad \{F_0\} \times \operatorname{Sym}^2 X.$$

The completion has volume $(16\pi^4)^4/24$. The open one-bubble stratum has volume $(16\pi^4)^3/2$. At $(F_0, x + y)$ with $x \neq y$, the tangent cone is $\mathbb{R}^8 \times (\mathbb{R}^4/\{\pm 1\})^2$ and its density is $1/4$. At $(F_0, 2x)$, its affine cone with Reeb action is $\mathfrak{M}_0(2,2)$, of complex dimension eight and density

11/108. There is no further $\mathbb{C}^2$ factor: the centre-of-mass coordinates already belong to $\mathfrak{M}_0(2,2)$.

Here $\chi(E) = r+s = 1$, so the moduli space is fine. Its nontrivial Uhlenbeck contraction and Corollary 5.3 give the fine example relevant to the extension assertion discussed in Section 5.

8.2. **A volume bound without a rational Kähler direction.** The potential argument used an integral polarization ray. Its exact volume formula nevertheless gives an upper bound for all hyperkähler metrics on the differentiable K3 surface.

Proposition 8.3. *Let $r \geq 2$, $\gcd(r,k) = 1$, and $N = rk - r^2 + 1 \geq 2$. For any hyperkähler metric g on a K3 surface, the moduli space of irreducible $SU(r)$ anti-self-dual connections of charge k satisfies*

$$\operatorname{Vol}(M_k^\circ(g)) \leq \frac{(16\pi^4 \operatorname{Vol}(X,g))^N}{N!}. \tag{38}$$

The volume of the empty moduli space is zero. If the positive three-plane of g contains a rational Kähler direction whose induced complex surface has Picard number one, equality holds whenever the moduli space is nonempty.

Proof. For a rational Kähler direction and Picard number one, the Mukai vector $(r, 0, r-k)$ is primitive. The polarization is Gieseker generic: a same-reduced-Hilbert-polynomial factor would have rank divisible by r because $\gcd(r,k) = 1$. The Kobayashi–Hitchin correspondence identifies the irreducible ASD locus with the slope-stable locally free locus. When nonempty, Theorem 5.1 and rescaling give equality.

Such positive three-planes are dense among positive three-planes in the K3 period domain: choose a rational positive vector close to one vector of the plane and choose the orthogonal positive two-plane generically to avoid every other integral hyperplane. The K3 period description gives nearby hyperkähler metrics of the same limiting volume. For any irreducible ASD connection, the hyperkähler Weitzenböck formula identifies its obstruction space with the tensor product of its parallel trace-free endomorphisms and the three-dimensional space of parallel self-dual forms. Irreducibility makes this space zero. The parameter-dependent implicit-function theorem therefore embeds each compact subset of $M_k^\circ(g)$ into the moduli spaces for nearby metrics, with smooth convergence of their L^2 tensors. Its volume is bounded by the total volumes of those nearby spaces, each at most the right side of (38). Exhaustion proves the inequality. □

This is a lower-semicontinuity argument for total moduli volume, not a completion theorem for an irrational positive three-plane. It does not assert that nonempty equality cases are dense. For instance, if $r \leq k < 2r$, then $\chi(E) = 2r - k > 0$; a slope-stable bundle of slope zero has neither sections nor dual sections, contradicting Riemann–Roch. The corresponding irreducible ASD moduli spaces are empty.

8.3. **A conditional expansion on a resolution.** The following elementary calculation records precisely what is needed for an exceptional-area expansion. Suppose $r = 2$, D is an exceptional divisor, and

$$q(\Lambda, D) = 0, \qquad q(D) = -2(N-1), \qquad D \cdot R = -2$$

for a contracted rational curve R. Assume also that $A_t = \Lambda - t[D]$ is Kähler for small $t > 0$. The area of R is $a = 2t$. Put $Q = q(\Lambda)/2 = 16\pi^4 V$. Fujiki's formula gives

$$\operatorname{Vol}(\Lambda) - \operatorname{Vol}(A_t) = \frac{Q^{N-1}}{4(N-2)!}\, a^2 + O(a^4). \tag{39}$$

Indeed $q(A_t)/2 = Q-(N-1)t^2$ and $\mathrm{Vol}(A_t) = (Q-(N-1)t^2)^N/N!$. When (31) applies, its coefficient is $\mathrm{Vol}(S_1)/4$. The displayed Beauville–Bogomolov identities and the Kähler-ray hypothesis are assumptions here. The fibre intersection $D \cdot R = -2$ alone does not prove either the square of D or the Kähler property of A_t.

8.4. **Further questions.** The continuous potential of Theorem 3.2 applies to polarized surfaces beyond K3. To obtain a global class and a Quillen metric there, one must separately control normalization, boundary codimension, determinant-line descent and the flat character. For a stratum inclusion, smoothness of a sheaf moduli space alone does not show that every elementary modification is a limit of bundles; this density condition must be established in the chosen component.

For K3 surfaces, several further metric questions remain. At collisions of bubble points, does the tangent metric equal the flat ADHM quotient metric without the extra uniqueness assertion (Q)? Do deeper strata with stable or reducible residual bundles have a totally geodesic metric structure? What rates accompany the Gromov–Hausdorff convergence near nonorbifold singularities? The explicit potential is continuous, but the present argument gives neither its optimal modulus of continuity nor a curvature-gluing expansion. Extending the exact volume and completion statements to all irrational Kähler directions, and treating nonprimitive vectors without a smooth symplectic resolution, requires further arguments rather than a change of notation in the theorems above.

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